

and integrating it into your application. Here's a detailed guide to help you through the process.

Step 1: Setting Up Your Development Environment

Before you start coding, you need to set up a suitable development environment. This involves installing necessary tools and libraries.

- 1. Install Python:** Ensure you have Python installed on your system. You can download the latest version from the official Python website.
- 2. Set Up a Virtual Environment:** Create a virtual environment to manage your project dependencies. Open your terminal and run:


```
python -m venv myenv
source myenv/bin/activate # On Windows,
use myenv\Scripts\activate
```
- 3. Install Required Libraries:** Install the necessary libraries for developing your ChatGPT plugin. This includes the OpenAI API client and Flask for creating a web interface:


```
pip install openai flask
```

Step 2: Getting Your OpenAI API Key

To interact with the ChatGPT model, you'll need an API key from OpenAI.

- 1. Sign Up and Get API Key:** Sign up at the OpenAI website and obtain your API key from the API section of your account.

Step 3: Creating the Plugin

Now that your environment is set up, let's start creating the ChatGPT plugin.

- 1. Initialize the Project:** Create a new directory for your project and navigate into it:


```
mkdir chatgpt_plugin
cd chatgpt_plugin
```
- 2. Create the Main Script:** Create a new Python file named `app.py`. This file will contain the code for your ChatGPT plugin.
- 3. Code the Plugin:** Open `app.py` in your favorite text editor and add the following code:


```
from flask import Flask, request, jsonify
import openai

app = Flask(__name__)
# Set your OpenAI API key
openai.api_key = 'your_openai_api_key_here'
@app.route('/chat', methods=['POST'])
def chat():
    data = request.json
    user_message = data.get('message', '')
    if not user_message:
        return jsonify({'error': 'No mes-
```

```
sage provided'}), 400
    try:
        response = openai.Completion.create(
            engine='text-davinci-003',
            prompt=user_message,
            max_tokens=150
        )
        ai_message = response.choices[0].
        text.strip()
        return jsonify({'response': ai_mes-
            sage})
    except Exception as e:
        return jsonify({'error': str(e)}),
500
if __name__ == '__main__':
    app.run(debug=True)
```

- 4. Run the Plugin:** Start your Flask server by running the following command in your terminal:

```
python app.py
```

This will start a local server on <http://127.0.0.1:5000>.

Step 4: Testing Your Plugin

With the server running, you can test your plugin using a tool like Postman or by writing a simple test script.

- 1. Using Postman:** Open Postman and create a new POST request to <http://127.0.0.1:5000/chat>. In the body, select raw and JSON, then enter:

```
{
    "message": "Hello, ChatGPT!"
}
```

Send the request and check the response from your plugin.

- 2. Using a Test Script:** Create a new Python file named `test_chat.py` and add the following code:

```
import requests

url = 'http://127.0.0.1:5000/chat'
data = {'message': 'Hello, ChatGPT!'}

response = requests.post(url, json=data)

if response.status_code == 200:
    print('Response:', response.json()
['response'])
else:
    print('Error:', response.json()
['error'])
```

Run the script:

```
python test_chat.py
```

This script will send a request to your plugin and print the AI's response.